Web Crawling and Basic Text Analysis

Many slides borrowed from Hongning Wang with modifications

Web Crawling

Web crawler

- An automatic program that browses the web for the purpose of Web content indexing and updating
  - Synonyms: spider, robot, bot

How does it work

- In pseudo code

```
Def Crawler(entry_point) {
        URL_list = [entry_point]
        while (len(URL_list)>0) {
                URL = URL_list.pop();
                if (isVisited(URL) or !isLegal(URL) or !checkRobotsTxt(URL))
                        continue;
                HTML = URL.open();
                for (anchor in HTML.listOfAnchors()) {
                        URL_list .append(anchor);
                }
                setVisited(URL);
                insertToIndex(HTML);
        }
}
```

Which page to visit next?

Is it visited already?
Or shall we visit it again?

Is the access granted?

Visiting strategy

- Breadth first
  - Uniformly explore from the entry page
  - Memorize all nodes on the previous level
  - As shown in pseudo code
- Depth first
  - Explore the web by branch
  - Biased crawling given the web is not a tree structure
- Focused crawling
  - Prioritize the new links by predefined strategies

Focused crawling

- Prioritize the visiting sequence of the web
  - The size of Web is too large for a crawler (even Google) to completely cover
  - In 1999, no search engine indexed more than 16% of the Web
  - In 2005, large-scale search engines index no more than 40-70% of the indexable Web.

Focused crawling

- Prioritize the visiting sequence of the web
  - The size of Web is too large for a crawler (even Google) to completely cover
  - Not all documents are equally important
  - Emphasize more on the high-quality documents
    - Maximize weighted coverage

$$\mathrm{WC}(t) = \sum_{p \in \mathcal{C}(t)} w(p)$$

Importance of page ***p***

Pages crawled till time ***t***

Weighted coverage till time ***t***

Focused crawling

- Prioritize by in-degree [Cho et al. WWW'98]
  - The page with the highest number of incoming hyperlinks from previously crawled pages is crawled next
- Prioritize by PageRank [Abiteboul et al. WWW'07, Cho and Uri VLDB'07]
  - Breadth-first in early stage, then compute/approximate PageRank periodically

Focused crawling

- Prioritize by topical relevance
  - In vertical search, only crawl relevant pages [De et al. WWW'94]
    - E.g., restaurant search engine should only crawl restaurant pages
  - Estimate the similarity to current page by anchor text or text near anchor [Hersovici et al. WWW'98]
  - User given taxonomy or topical classifier [Chakrabarti et al. WWW'98]

Avoid duplicate visit

- Given web is a graph rather than a tree, avoid loop in crawling is important
- How to check
  - trie or hash table
- What to check
  - URL: must be normalized, not necessarily can avoid all duplication
    - http://dl.acm.org/event.cfm?id=RE160&CFID=516168213&CFTOKEN=99036335
    - http://dl.acm.org/event.cfm?id=RE160
  - Page: minor change might cause misfire
    - Timestamp, data center ID change in HTML

Politeness policy

- Crawlers can retrieve data much quicker and in greater depth than human searchers
- Costs of using Web crawlers
  - Network resources
  - Server overload
- Robots exclusion protocol
  - a text file called robots.txt in the root of the web site hierarchy, requests specified robots to ignore specified files or directories when crawling a site

Robot exclusion protocol examples

- Exclude specific directories:

```
User-agent: *
 Disallow: /tmp/
 Disallow: /cgi-bin/
 Disallow: /users/paranoid/
```

- Exclude a specific robot:

```
User-agent: GoogleBot
 Disallow: /
```

- Allow a specific robot:

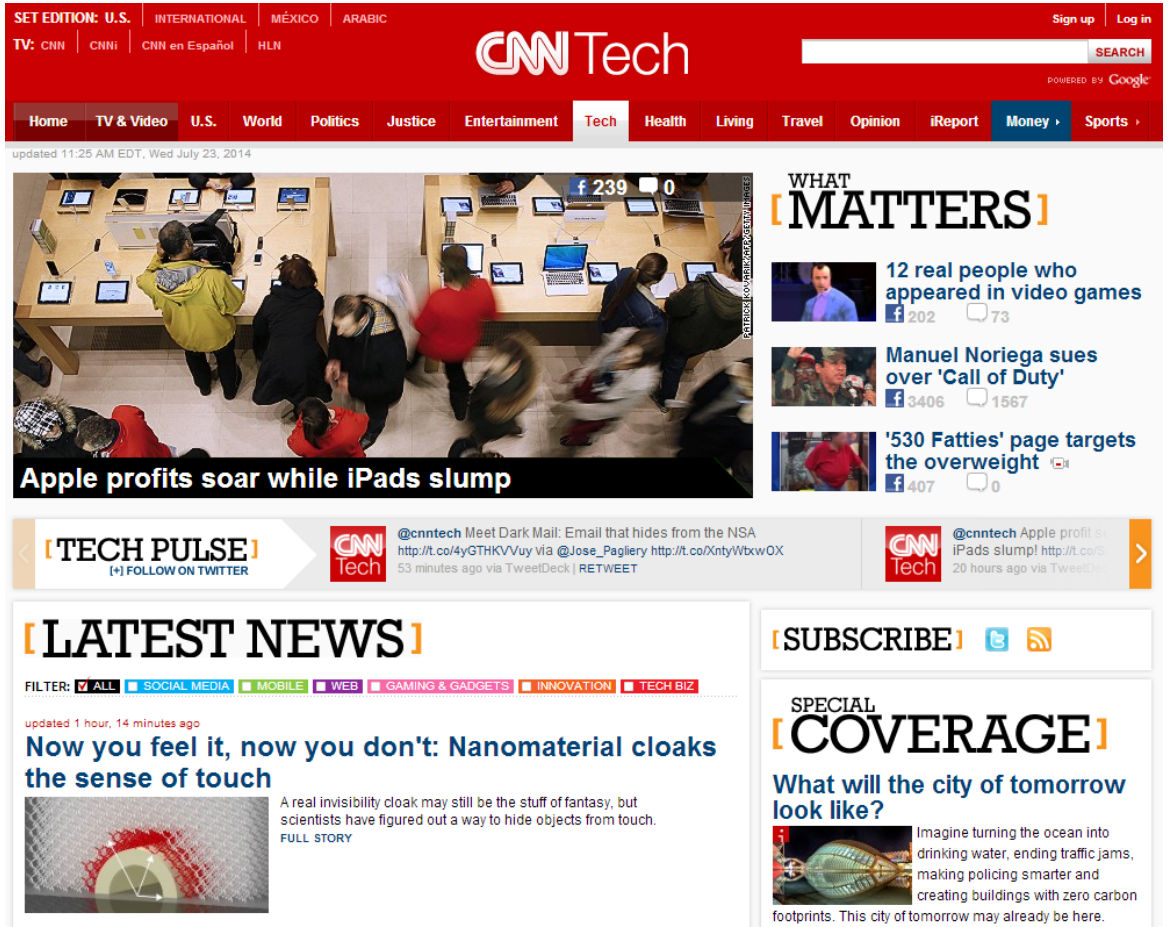
```
User-agent: GoogleBot
 Disallow:

 User-agent: *
 Disallow: /
```

Basic Text Analysis

Analyze crawled web pages

- What you care from the crawled web pages



Analyze crawled web pages

- What machine gets from the crawled web pages

```
<!DOCTYPE HTML>
<html lang="en-US">
<head>
<title>Technology News - Computers, Internet, Invention and Innovation Tech from CNN.com</title>
<meta http-equiv="content-type" content="text/html;charset=utf-8"/>
<meta http-equiv="last-modified" content="2014-07-23T15:25:56Z"/>
<meta name="robots" content="index,follow"/>
<meta name="googlebot" content="noarchive"/>
<meta name="viewport" content="width=1024"/>
<meta name="title" content="Technology News - Computers, Internet, Invention and Innovation Tech from CNN.com"/>
<meta name="description" content="Find information about the latest advances in technology at CNN. CNN Technology news and video covers the internet,
business and personal tech, video games, and more."/>
<meta name="keywords" content="CNN, CNN news, CNN.com, CNN TV, news, news online, breaking news, U.S. news, world news, weather, business, CNN Money,
sports, politics, law, technology, entertainment, education, travel, health, special reports, autos, developing story, news video, CNN Intl"/>
<link rel="canonical" href="http://www.cnn.com/TECH/"/>
<link type="image/png" rel="apple-touch-icon" href="http://i.cdn.turner.com/cnn/.e/img/3.0/global/misc/apple-touch-icon.png"/>
<link type="application/rss+xml" rel="alternate" href="http://rss.cnn.com/rss/cnn_tech.rss" title="CNN - Tech [RSS]"/>
<link type="application/rss+xml" rel="alternate" href="http://rss.cnn.com/rss/cnn_topstories.rss" title="CNN - Top Stories [RSS]"/>
<link type="application/rss+xml" rel="alternate" href="http://rss.cnn.com/rss/cnn_latest.rss" title="CNN - Recent Stories [RSS]"/>
<link type="application/opensearchdescription+xml" rel="search" href="/tools/search/cnncom.xml" title="CNN.com"/>
<link type="application/opensearchdescription+xml" rel="search" href="/tools/search/cnncomvideo.xml" title="CNN.com Video"/>
<link href="https://plus.google.com/u/0/b/117515799321987910349/117515799321987910349/posts" rel="publisher"/>
<link type="text/css" rel="stylesheet" href="http://z.cdn.turner.com/cnn/tmpl_asset/static/www_section/2695/css/techlib-min.css"/>
<script>
var cnnCVPAdSection='cnn.com_technology_section_homepage',
cnnIsSectionPage=true,
cnnSectionName='Tech',
cnnSectionFront='Tech',
sectionName='tech';
</script>
<script src="http://z.cdn.turner.com/cnn/tmpl_asset/static/www_section/2695/js/techlib-min.js"></script>
<script>
var cnnPageType="Section";
if(typeof(cnn_metadata)=='undefined'){var cnn_metadata={};}
var cnn_edtnswtchver='www';
cnn_metadata.section=['tech','tch : frontpage'];
cnn_metadata.friendly_name='Tech Home Page';
cnn_metadata.template_type='section front';
var CNN_gallery_0_ad_0="/cnn_adspaces/3.0/technology/main/bot1.120x90.ad";
var CNN_gallery_0_ad_1="/cnn_adspaces/3.0/technology/main/bot2.120x90.ad";
var CNN_gallery_0_ad_2="/cnn_adspaces/3.0/technology/main/bot3.120x90.ad";
</script>
```

Basic text analysis techniques

- Need to analyze and index the crawled web pages
  - Extract informative content from HTML
  - Build machine accessible data <u>representation</u>

HTML parsing

- Generally difficult due to the free style of HTML
- Solutions
  - Shallow parsing
    - Remove all HTML tags
    - Only keep text between <title></title> and
  - Automatic wrapper generation [Crescenzi et al. VLDB'01]
    - Wrapper: regular expression for HTML tags' combination
    - Inductive reasoning from examples
  - Visual parsing [Yang and Zhang DAR'01]
    - Frequent pattern mining of visually similar HTML blocks

HTML parsing

- jsoup
  - Java-based HTML parser
    - scrape and parse HTML from a URL, file, or string to DOM (Document Object Model) tree
    - Find and extract data, using DOM traversal or CSS selectors
      - children(), parent(), siblingElements()
      - getElementsByClass(), getElementsByAttributeValue()
  - Python version: Beautiful Soup

How to represent a document

- Represent by a string?

<HEAD>Crowds in Liverpool to Mark 10th Anniversary of John Lennon's Death</HEAD>
<DATELINE>LIVERPOOL, England (AP) </DATELINE>
<TEXT>
Dozens of fans of rock legend and former Beatle John Lennon gathered in the snow on a windy Saturday for a ceremony marking the 10th anniversary of his death. Liverpool's mayor, Dorothy Gavin, led Lennon devotees who laid wreaths at the foot of a bronze statue of The Beatles in the city's Cavern Walks shopping center. The center was built on the original site of the Cavern Club, made famous when The Beatles played there in the 1960s, and has become a place of pilgrimage. ``Give peace a chance,'' the title of one of singer-songwriter Lennon's greatest hits, was the theme for the day.
…
Lennon and his wife, Yoko Ono, were returning to their apartment in New York's Dakota apartment building after a recording session on Dec. 8, 1980, when Lennon was shot to death by Mark David Chapman, a deranged fan to whom Lennon had given his autograph only hours before. Lennon was 40. A spokesman for the Lennon family said Ms. Ono and the couple's son, Sean, were in Europe and would spend the anniversary privately.
…
Peebles said late in 1980 that Lennon had just recovered from a period when he had ``gone off the rails'' and his relationship with Ms. Ono had suffered. ``But (when I saw him) they'd had the baby, Sean had been born, and everything was great.''
</TEXT>

- – Bag-of-Words representation!

Full text indexing

- Bag-of-Words representation
  - Doc1: Information retrieval is helpful for everyone.
  - Doc2: Helpful information is retrieved for you.

	information	retrieval	retrieved	is	helpful	for	you	everyone
Doc1	1	1	0	1	1	1	0	1
Doc2	1	0	1	1	1	1	1	0

Word-document adjacency matrix

Full text indexing

- Bag-of-Words representation
  - Assumption: word is independent from each other
  - Pros: simple
  - Cons: grammar and order are missing
  - ***The most frequently used document representation***

Tokenization

- Break a stream of text into meaningful units
  - Tokens: words, phrases, symbols
    - **Input:** It’s not straight-forward to perform so-called “tokenization.”
    - **Output(1):** 'It’s', 'not', 'straight-forward', 'to', 'perform', 'so-called', '“tokenization.”'
    - **Output(2):** 'It', ’’', 's', 'not', 'straight', '-', 'forward, 'to', 'perform', 'so', '-', 'called', ‘“', 'tokenization', '.', ’”‘
  - Definition depends on language, corpus, or even context

Tokenization

- Solutions
  - Regular expression
    - [\w]+: so-called -> ‘so’, ‘called’
    - [\S]+: It’s -> ‘It’s’ instead of ‘It’, ‘’s’
  - Statistical methods
    - Explore rich features to decide where is the boundary of a word
      - Apache OpenNLP (http://opennlp.apache.org/)
      - Stanford NLP Parser (http://nlp.stanford.edu/software/lex-parser.shtml)
    - Online Demo
      - Stanford (http://nlp.stanford.edu:8080/parser/index.jsp)
      - UIUC (http://cogcomp.cs.illinois.edu/curator/demo/index.html)

Full text indexing

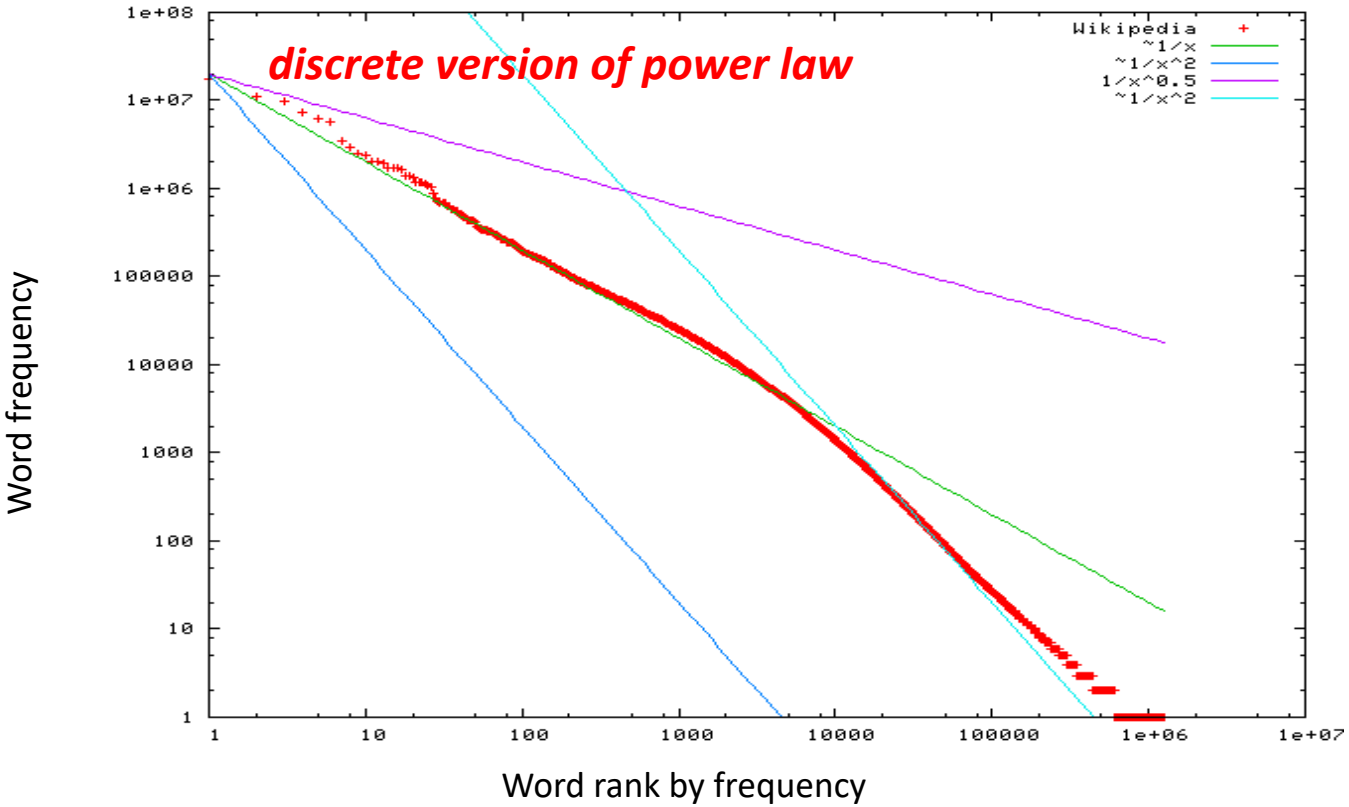
- Index document with all the occurring word
  - Pros
    - Preserve all information in the text (hopefully)
    - Fully automatic
  - Cons
    - Vocabulary gap: cars v.s., car
    - Large storage
  - Solution
    - Construct controlled vocabulary

Statistical property of language

- Zipf's law
  - Frequency of any word is inversely proportional to its rank in the frequency table
  - Formally
    - $f(k; s, N) = \frac{1/k^s}{\sum_{n=1}^{N} 1/n^s}$

      where k is rank of the word; N is the vocabulary size; s is a language-specific parameter

Statistical property of language



A plot of word frequency in Wikipedia (Nov 27, 2006)

*In the Brown Corpus of American English text, the word "the" is the most frequently occurring word, and by itself accounts for nearly **7%** of all word occurrences; the second-place word "of" accounts for slightly over **3.5%** of words.*

Zipf's law tells us

- Head words may take large portion of occurrence, but they are semantically meaningless
  - E.g., the, a, an, we, do, to
- Tail words take major portion of vocabulary, but they rarely occur in documents
  - E.g., dextrosinistral
- The rest is most representative
  - To be included in the controlled vocabulary

Automatic text indexing

Remove non-informative words

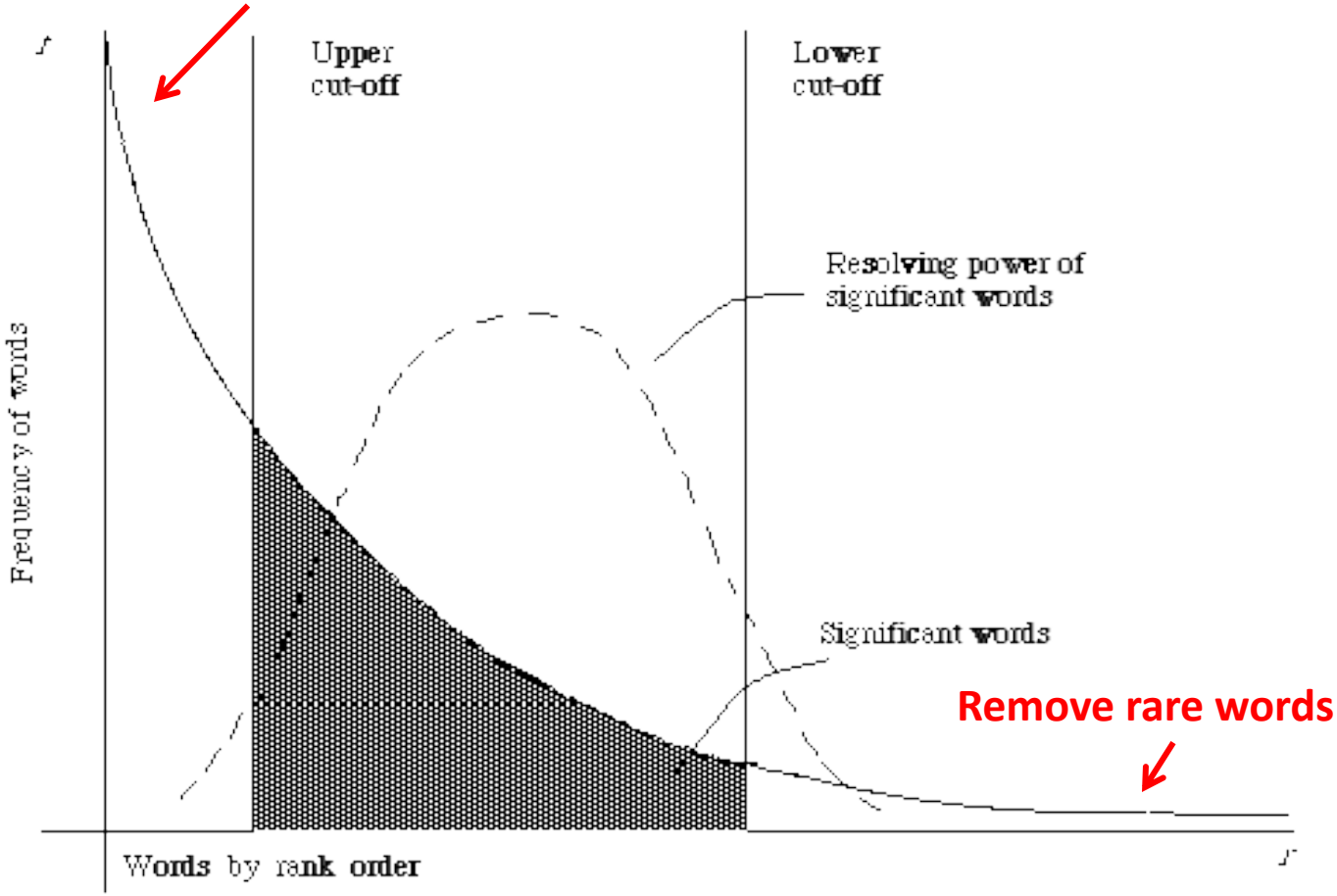


Figure 2.1. A plot of the hyperbolic curve relating f, the frequency of occurrence and r, the rank order (Adapted from Schultz[44] page 120)

Normalization

- Convert different forms of a word to normalized form in the vocabulary
  - U.S.A -> USA, St. Louis -> Saint Louis
- Solution
  - Rule-based
    - Delete periods and hyphens
    - All in lower case
  - Dictionary-based
    - Construct equivalent class
      - Car -> “automobile, vehicle”
      - Mobile phone -> “cellphone”

Stemming

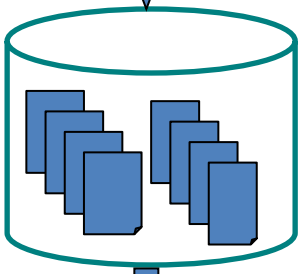
- Reduce inflected or derived words to their root form
  - Plurals, adverbs, inflected word forms
    - E.g., automate(s), automatic, automation => automat
  - Bridge the vocabulary gap
  - Solutions (for English)
    - Porter stemmer: pattern of vowel-consonant sequence
    - Krovetz Stemmer: morphological rules

Abstraction of search engine architecture

Indexed corpus

Crawler

1. Visiting strategy
2. Avoid duplicated visit
3. Re-visit policy



Doc Analyzer

1. HTML parsing
2. Tokenization
3. Stemming/normalization
4. Stopword/controlled vocabulary filter

Doc Representation

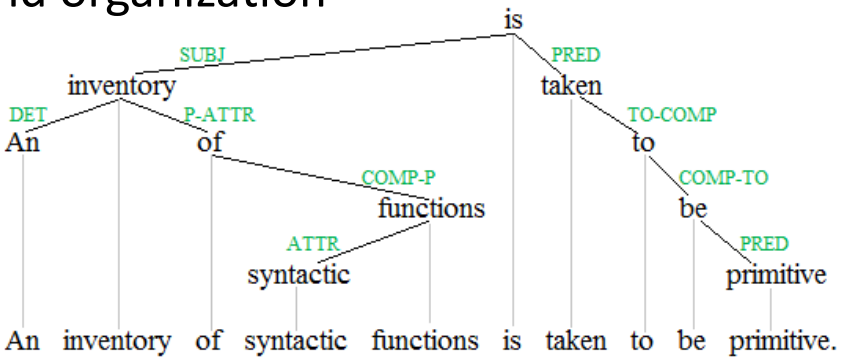
Terms	Documents													
	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12	M13	M14
abnormalities	0	0	0	0	0	0	0	1	0	1	0	0	0	0
age	1	0	0	0	0	0	0	0	0	0	0	1	0	0
behavior	0	0	0	0	1	1	0	0	0	0	0	0	0	0
blood	0	0	0	0	0	0	0	1	0	0	1	0	0	0
close	0	0	0	0	0	0	1	0	0	0	1	0	0	0
culture	1	1	0	0	0	0	0	1	1	0	0	0	0	0
depressed	1	0	1	1	1	0	0	0	0	0	0	0	0	0
discharge	1	1	0	0	0	1	0	0	0	0	0	0	0	0
disease	0	0	0	0	0	0	0	0	1	0	1	0	0	0
fast	0	0	0	0	0	0	0	0	0	1	0	1	1	1
generation	0	0	0	0	0	0	0	0	1	0	0	0	1	0
oestrogen	0	0	1	1	0	0	0	0	0	0	0	0	0	0
patients	1	1	0	1	0	0	0	1	0	0	0	0	0	0
pressure	0	0	0	0	0	0	0	0	0	0	1	0	0	1
rats	0	0	0	0	0	0	0	0	0	0	0	0	1	1
respect	0	0	0	0	0	0	0	1	0	0	0	1	0	0
rise	0	0	0	1	0	0	0	0	0	0	0	0	0	1
study	1	0	1	0	0	0	0	0	1	0	0	0	0	0

BagOfWord representation!

Automatic text indexing

Query: "to be or not to be"

- In modern search engine
  - **No** stemming or stopword removal, since computation and storage are no longer the major concern
  - More advanced NLP techniques are applied
    - Named entity recognition
      - E.g., people, location and organization
    - Dependency parsing



What you should know

- Basic techniques for crawling
- Zipf’s law
- Basic text analysis techniques
- Bag-of-Words document representation

Chapter Reading

- Introduction to Information Retrieval
  - Chapter 20: Web crawling and indexes
    - Section 20.1, Overview
    - Section 20.2, Crawling
  - Chapter 2: The term vocabulary and postings lists
    - Section 2.2, Determining the vocabulary of terms
  - Chapter 5: Index compression
    - Section 5.1, Statistical properties of terms in information retrieval

Reference I

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Reference II

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